

Suggested Roles

Role	Main Tasks
Project Manager	Organizes workflow, tracks progress, keeps documentation
Lead Programmer	Handles main game loop, movement, and logic
Graphics Developer	Draws sprites (skier, trees, rocks, snow)
UI & Scoring Developer	Implements score, distance, and menus
Tester & Debugger	Tests gameplay, fixes bugs, maintains version backups

16-Period Timeline

Phase 1: Planning & Setup (Periods 1–3)

Goal: Understand project, divide roles, and start basic setup.

Period	Tasks	Deliverable
1	- Play original SkiFree for reference - Discuss features (movement, obstacles, scoring, optional Yeti) - Assign group roles	Project plan & team roles
2	- Set up Processing project - Create main sketch (<code>setup()</code> & <code>draw()</code> functions) - Sketch interface (window size, background color)	Running blank window
3	- Write pseudocode for core gameplay (loop, controls, collision) - Create folder for assets (images, sounds) - Start simple design document	Game design outline

Phase 2: Core Game Mechanics (Periods 4–8)

Goal: Build the main skiing experience — movement, background, and collisions.

Period	Tasks	Deliverable
4	- Draw skier (simple triangle or rectangle first) - Add keyboard input (left/right movement, speed control)	Skier moves on screen
5	- Create scrolling background (simulate downhill movement) - Test skier staying centered while background scrolls	Smooth scrolling effect

6	- Add obstacles (trees, rocks) using arrays or ArrayList - Randomly generate obstacle positions	Obstacles appear on screen
7	- Implement collision detection (bounding boxes or distance checks) - Add crash/reset logic	Collisions work
8	- Add scoring system (based on distance or time survived) - Display score on screen	Working scoring HUD

Phase 3: Visuals, Sound, and Extras (Periods 9–12)

Goal: Make the game look and feel more like SkiFree.

Period	Tasks	Deliverable
9	- Replace placeholder graphics with sprites (skier, trees, snow textures) - Test scaling and image movement	Polished visuals
10	- Add menus (start, pause, restart) - Add text for “Game Over” or “Press space to restart”	Functional menu system
11	- Add sound effects (crash, skiing noise) using <code>minim</code> or built-in Processing sound library	Sound integration
12	- Add optional fun features (Yeti chase, jumps, speed zones, etc.)	Bonus mechanics

Phase 4: Testing, Polish & Presentation (Periods 13–16)

Goal: Refine gameplay, fix bugs, and prepare for final presentation.

Period	Tasks	Deliverable
13	- Full gameplay testing - Log and fix major bugs	Stable game build
14	- Polish visuals (UI alignment, text size, smooth transitions) - Optimize code & frame rate	Final game version
15	- Prepare presentation slides - Write documentation (features, how it works, who did what)	Completed documentation
16	- Final demo & presentation to class - Show gameplay and team roles	Project showcase
